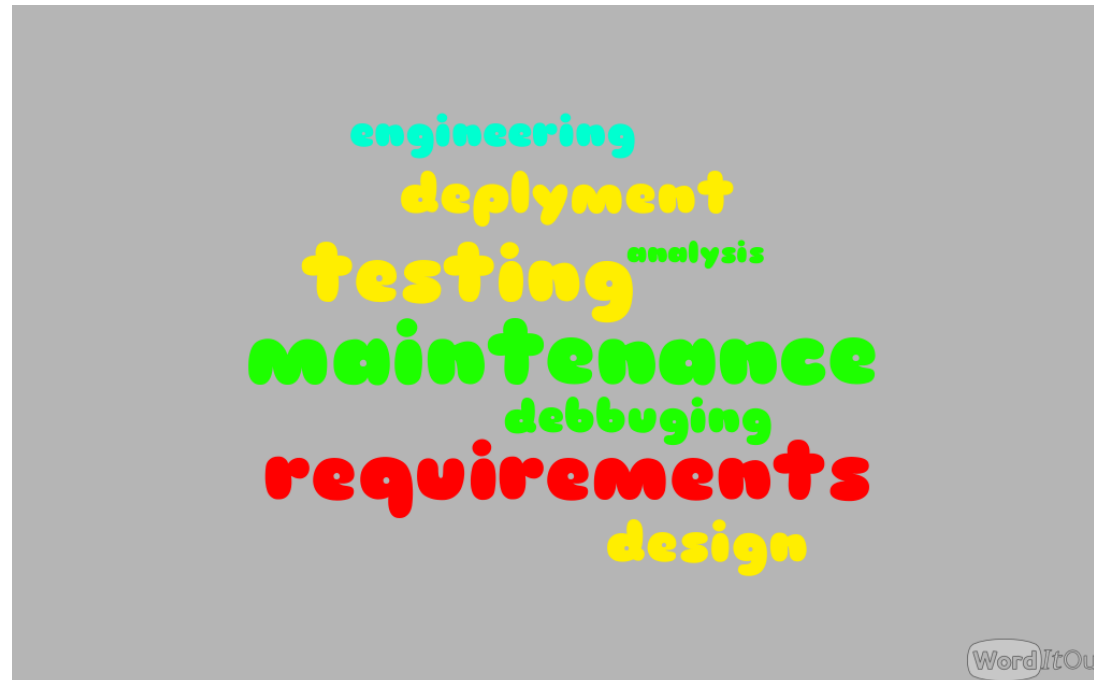


Course 9

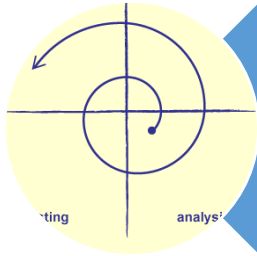
Software process quality

Software development processes

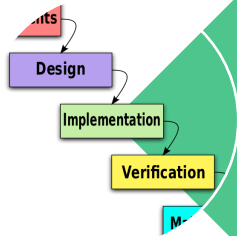
- Software process = set of related activities that lead to the production of software



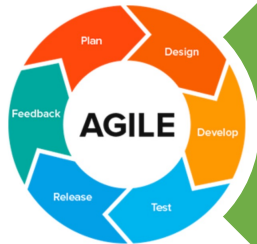
Software development methodologies



Spiral



Waterfall



Agile

Pro

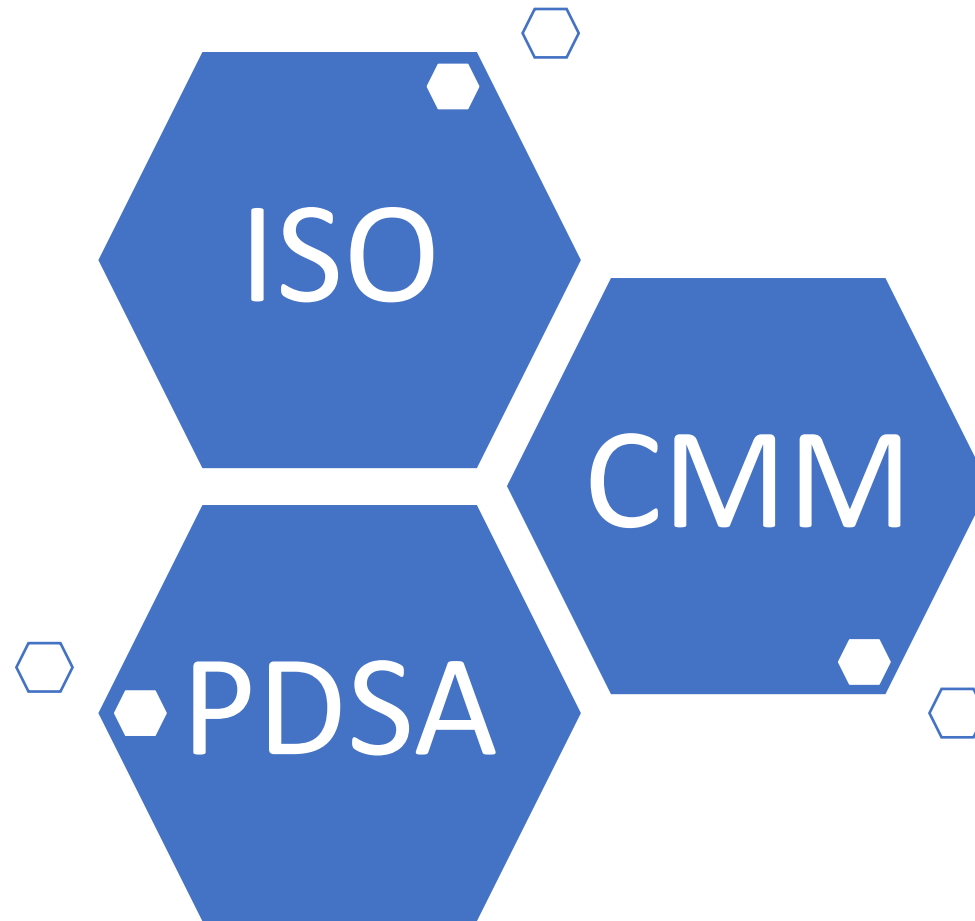
- Estimation
- Deliver
- Control



Cons

- Success = people + technology + manager
- Introduce bureaucracy
- Introduce extra costs
- Only for large projects and companies

Achieve quality processes / improve processes



ISO

- WHY:
 - Software development and maintenance – highest quality
 - Better communication and understanding between:
 - Development & maintenance
 - Developers & external contributors
 - Providers & customers
 - Better management of complex projects

ISO 9000

- **ISO 9000**: requirements a quality system must meet; addresses quality management and quality assurance standards. NOT how
- **ISO 9001**: what quality standards should be followed. NOT how
- **ISO 9000:2000**: serie - 8 key principles: Customer Focus, Leadership, Involvement of People, Process Approach, System Approach to Management, Continual improvement, Factual Approach to Decision Making and Mutually Beneficial Supplier Relationships

ISO 9000 documentation

Level 1: approach and responsibilities – Quality Manual

Level 2: who, what, when - Procedures

Level 3: how - Work / job instructions

Level 4: results – Records, documentation

ISO 9001: 2000 Structure



Procedure

- Certification based



- Valid 3 years
- Requires a working system of Quality Management (regular internal audits, external audits)

CMM

CMM (Capacity – Maturity Model)

CMMI (CMM Integration)

- Developed by SEI (Software Engineering Institute) from Carnegie Mellon Univ.
- Maturity = degree of formality to:
 - Define steps
 - Manage result metrics
 - Optimize processes

CMMI terms

1. Maturity levels – increased performance
2. KPA (Key Process Area): identify & group activities to achieve goals
3. Goals: scope + intend + range for each KPA
 - Indicates *capability* of org. for a *maturity* level
4. Common features: practices assoc. to a KPA
5. Key practices: implementation and adoption of KPA

Process Areas

A red banner with a wavy, ribbon-like border, containing the text "Over 500 pages" in white.

Over 500 pages

- Requirements Management
- Organizational Process Definition
- Project Planning
- Organizational Training
- Project Monitoring & Control
- Integrated Project Management
- Supplier Agreement Management
- Risk Management
- Measurement & Analysis
- Integrated Teaming
- Process & Product Quality Assurance
- Integrated Supplier Management
- Configuration Management
- Decision Analysis & Resolution
- Requirements Development
- Organizational Environment for Integration
- Technical Solution
- Organizational Process Performance
- Product Integration
- Quantitative Project Management
- Verification
- Organizational Innovation & Deployment
- Validation
- Causal Analysis & Resolution
- Organizational Process Focus

Maturity levels & KPAs

Level 1

- Initial

Level 2

- Repeatable

Level 3

- Defined

Level 4

- Managed

Level 5

- Optimizing

CMM Maturity Level 1: Initial

- No project management in place
- Ad hoc practices
- Activities are not preplanned – not pro-active, but reactive - respond to crisis
- Unpredictable process
- Cannot predict time and cost of development

Initiate project management

CMM Maturity Level 2: Repeatable

- Basic SE management practices
- Planning and management are based on experience (repeatable)
- Track costs and schedules
- Identify problems as arise and take immediate corrective action

Standardized process:

- *Peer review*
- *Software product engineering: methods, technologies*
- *Integrated software management*
- *Organization-level: training, process definition*

CMM Maturity Level 3: Defined

- Process for development fully documented
- Reviews used to achieve software quality
- Introduce CASE Tools

- *Software quality management*
- *Quantitative process management:*
 - Statistical process management: quantify quality and cost parameters
 - Assess relative quality of each product

CMM Maturity Level 4: Managed

- Organization sets quality and productivity goals for each project
- Continuous measurement
- Statistical quality control in place

- *Automatically collect process data*
- *use data to analyze and improve*
- *Defect prevention*

CMM Maturity Level 5: Optimizing

- Statistical quality and process control techniques
- Process has positive feedback loop
- To maintain level: Continue improvement and optimization of the process

2 representations of CMMI

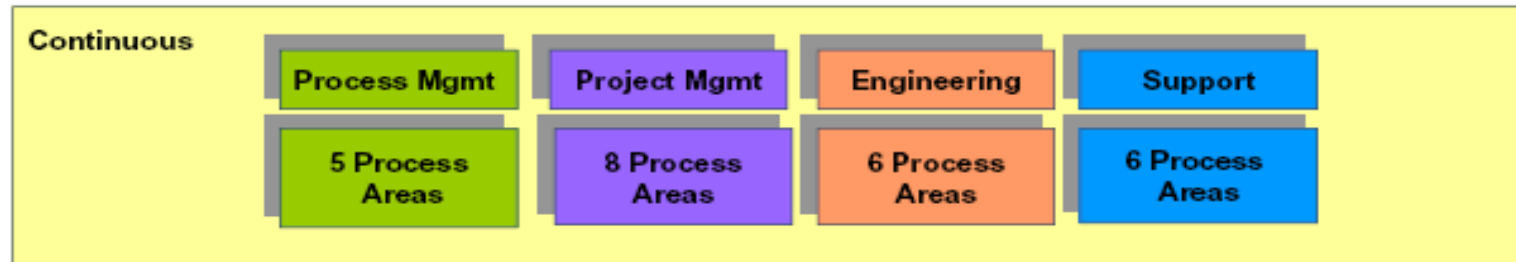
Continuous

- Allows to select the order of changes
- Predefined sets of PA to define improvement
- Focuses on improvements within PA

Staged

- Given sequence of improvements
- Allows to select PA and improve it
- Focuses on organization improvement

Continuous View of CMMI



- Organizational Process Focus
- Organizational Process Definition
- Organizational Training
- Organizational Process Performance
- Organizational Innovation and Deployment

- Requirements Management
- Requirements Development
- Technical Solution
- Product Integration
- Verification
- Validation

- Project Planning
- Project Monitoring and Control
- Supplier Agreement Management
- Integrated Project Management
- Risk Management
- Integrated Teaming
- Integrated Supplier Management
- Quantitative Project Management

- Configuration Management
- Process and Product Quality Assurance
- Measurement and Analysis
- Decision Analysis and Resolution
- Organizational Environment for Integration
- Causal Analysis and Resolution

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Capability Levels

Incomplete

Performed

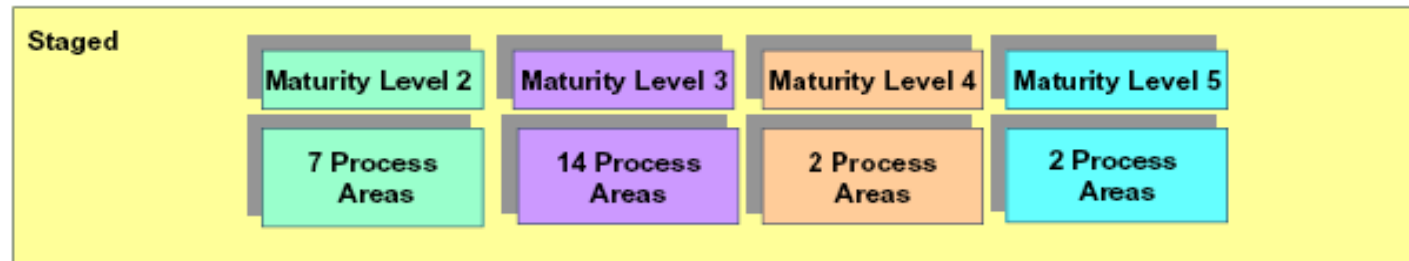
Managed

Defined

Quantitatively Managed

Optimizing

Staged View of CMMI



- Requirements Management
- Project Planning
- Project Monitoring and Control
- Supplier Agreement Management
- Measurement and Analysis
- Process and Product Quality Assurance
- Configuration Management

- Requirements Development
- Technical Solution
- Product Integration
- Verification
- Validation
- Organizational Process Focus
- Organizational Process Definition

- Organizational Training
- Integrated Project Management
- Risk Management
- Integrated Teaming
- Integrated Supplier Management
- Decision Analysis and Resolution
- Organizational Environment for Integration

- Organizational Process Performance
- Quantitative Project Management

- Organizational Innovation and Deployment
- Causal Analysis and Resolution

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Capability Levels

Initial

Managed

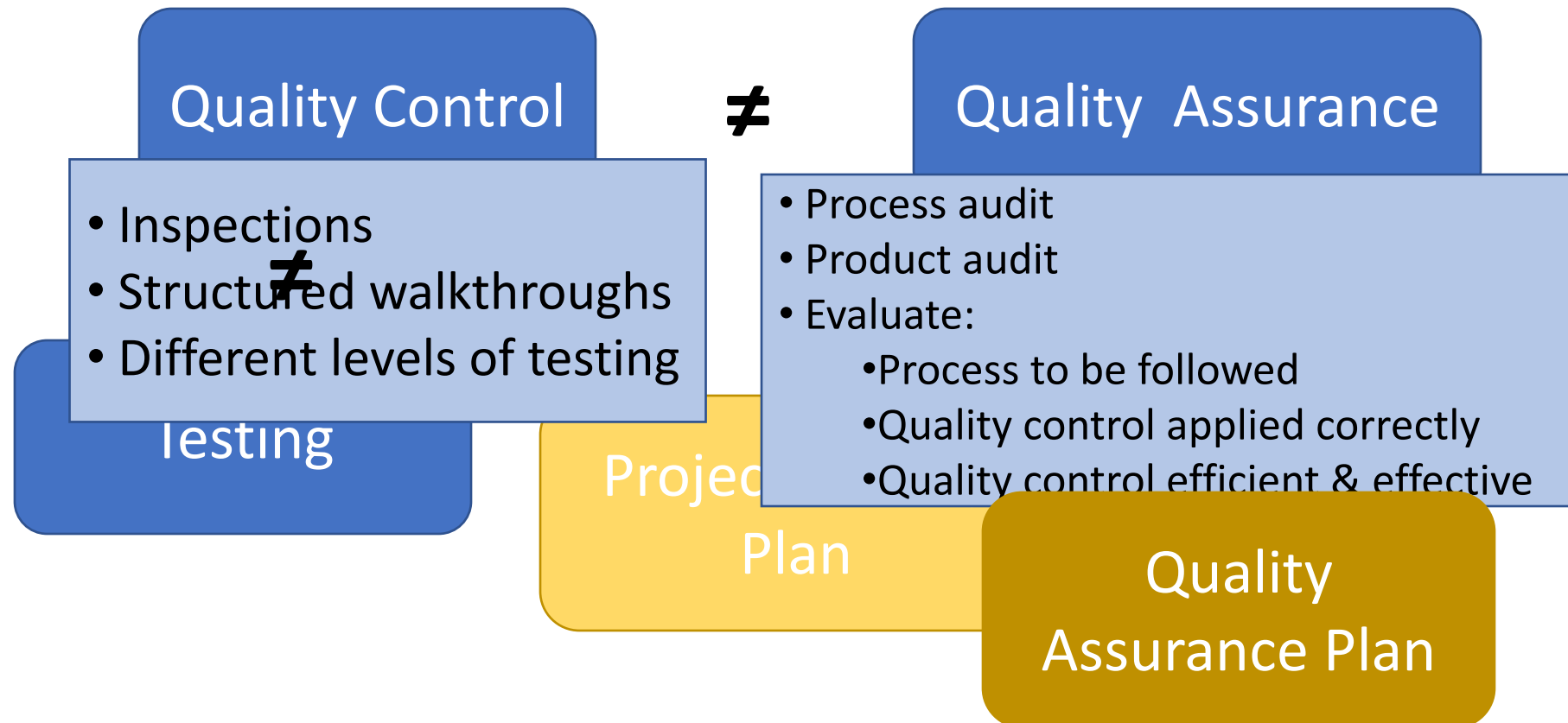
Defined

Quantitatively
managed

Optimizing

PPQA (Process and product Quality Assurance)

- Project management OR quality management



PPQA in CMMI

- Is a Process Area (PA) - components

- Required

- What an organization must achieve to satisfy a process area
- Specific + generic goals

- Expected

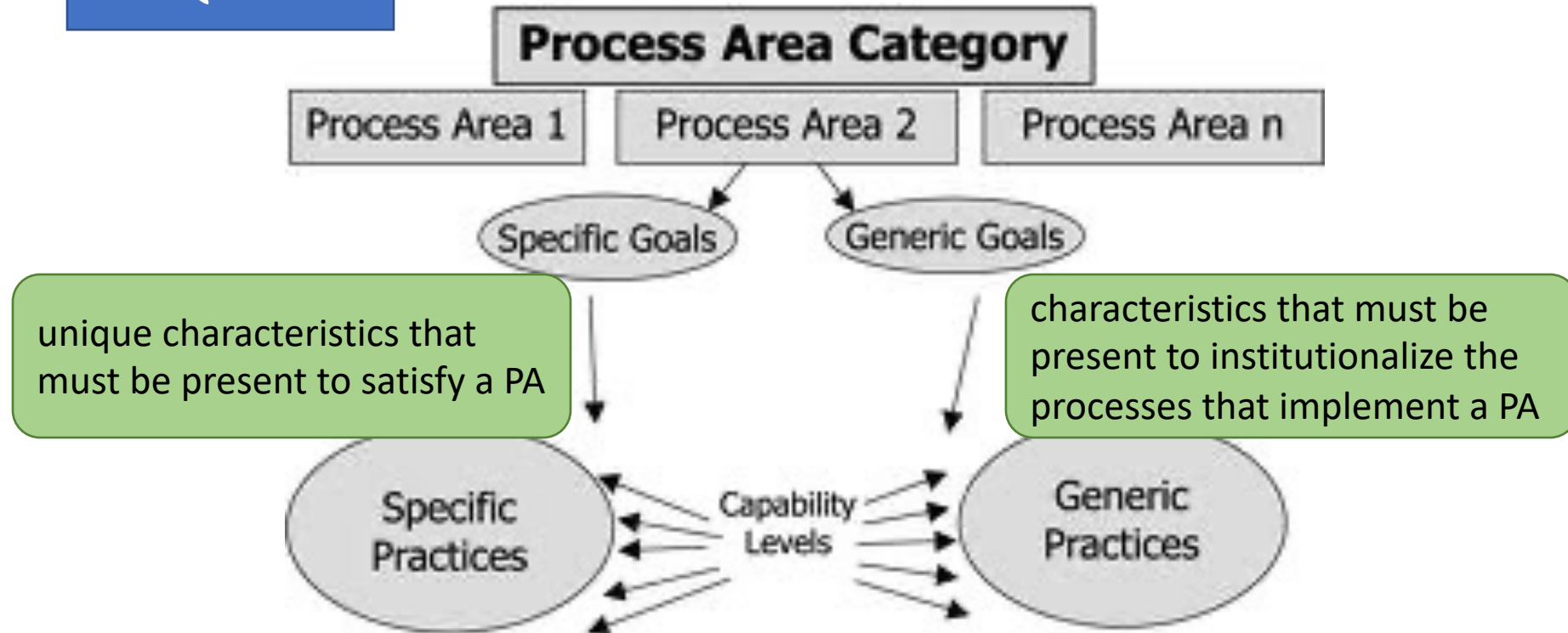
- what an organization may implement to achieve a required component
- Specific + generic practices

- Informative

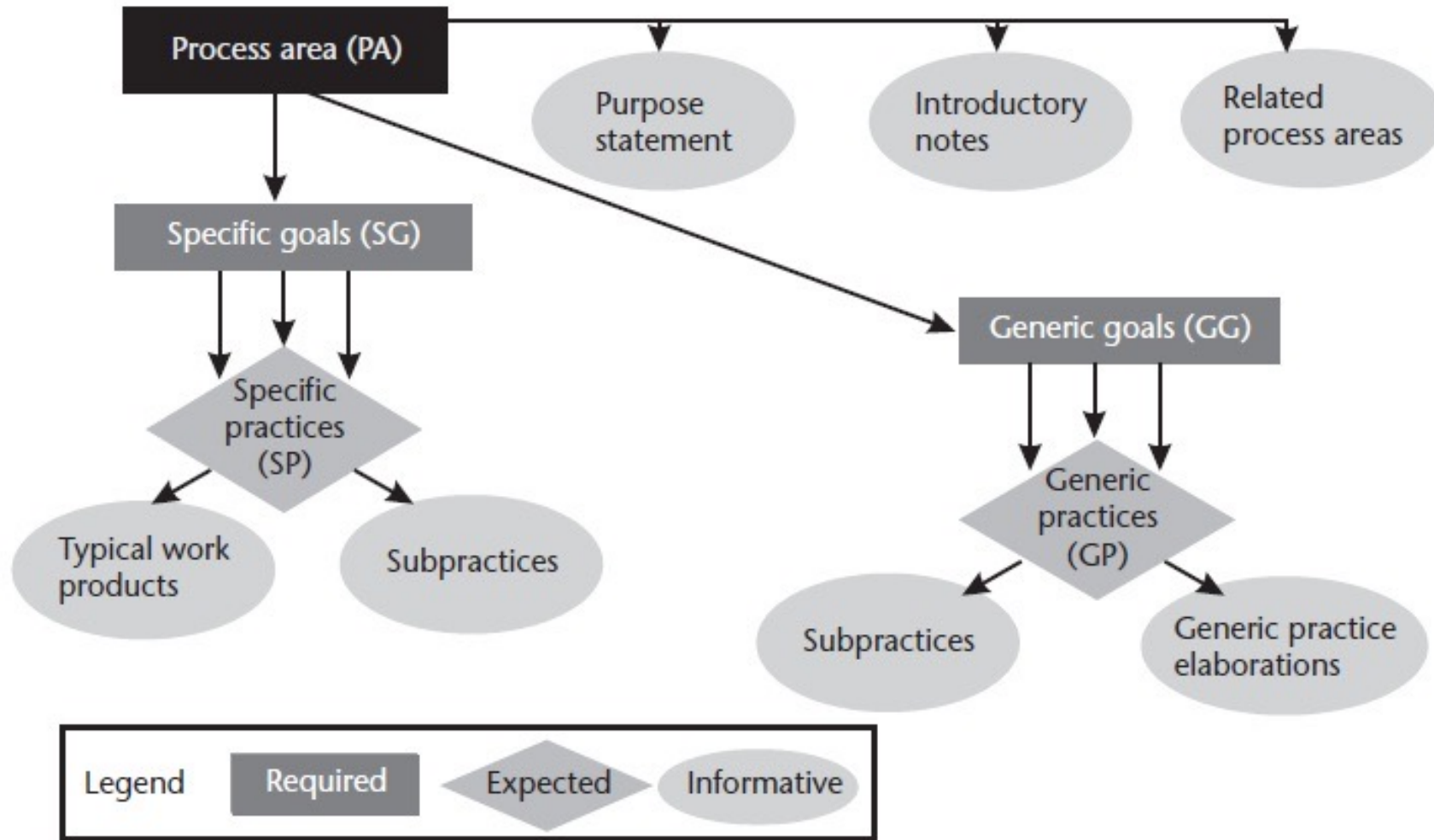
- how to approach the required and expected components

PPQA in CMMI

PPQA = PA



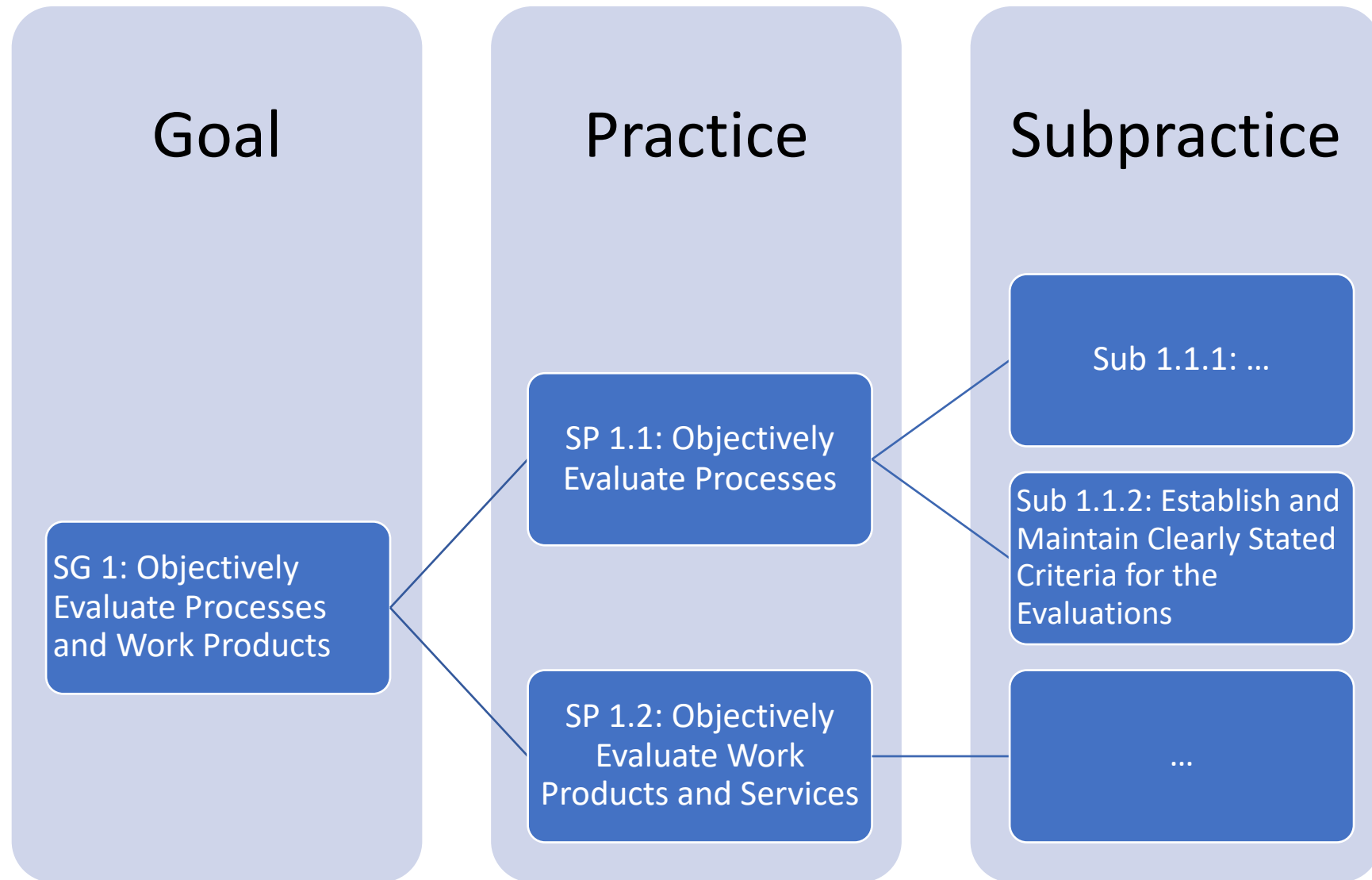
PPQA in CMMI



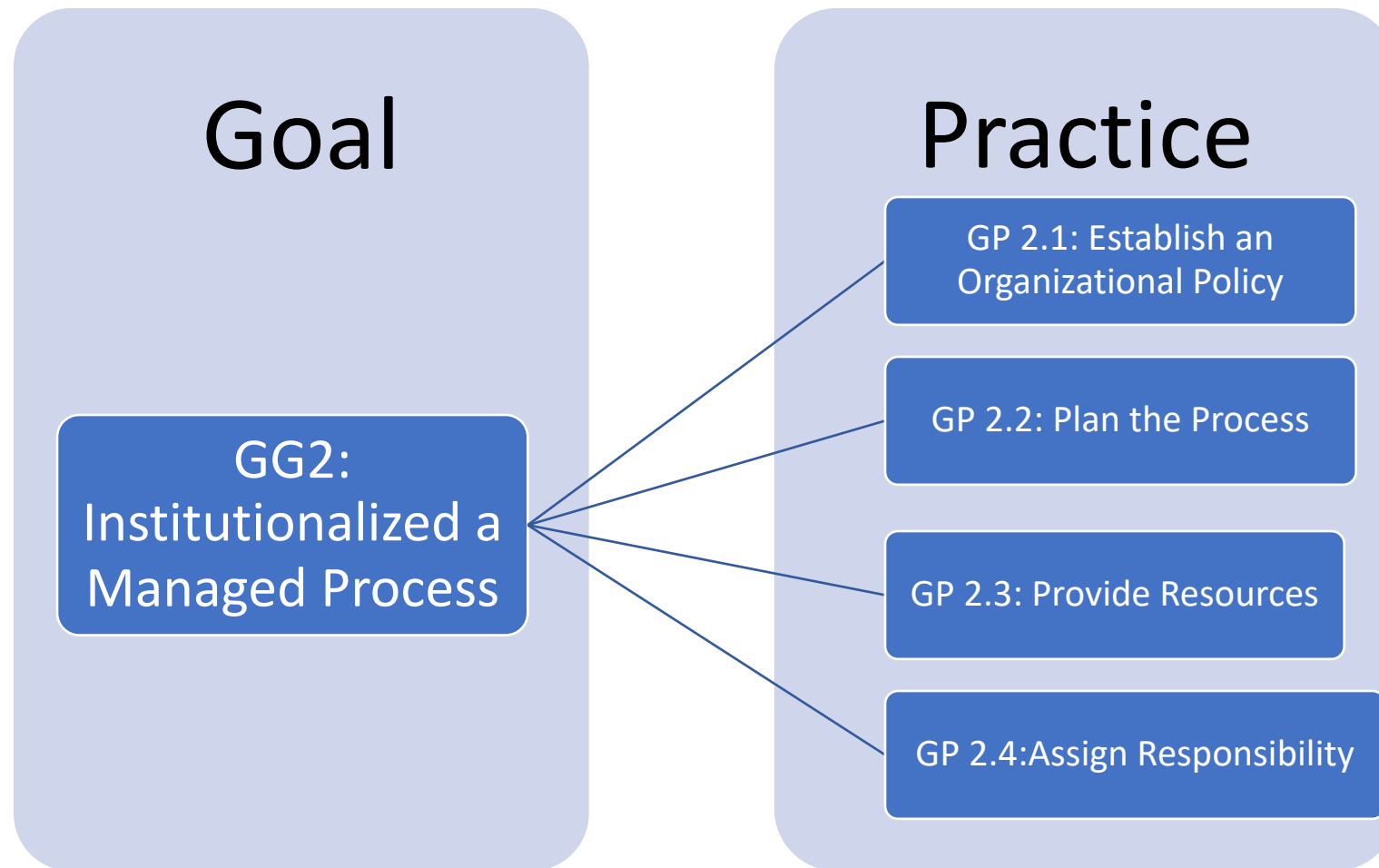
Specific goals & practices

- **SG1—Objectively Evaluate Processes and Work Products:**
 - Objectively Evaluate Processes;
 - Objectively Evaluate Work Products and Services.
- **SG2—Provide Objective Insight:**
 - Communicate and Ensure Resolution of Noncompliance Issues;
 - Establish Records.





Generic goals & practices



CMMI Appraisal

- Awarded a maturity level rating: 1-5
- 3 classes: A, B, C
- [Standard CMMI Appraisal Method for Process Improvement](#) (SCAMPI)

Success stories

- Boeing's Space Transportation Systems Software
 - Tata Consultancy Services (TCS)
 - Thales ATM
 - CMMI level 5:
 - Romania 0
 - Germany 0
 - France 1
 - UK 3
- A substantial shift in defect detection, from 89% late detection by testing to **83% early** detection by application of various review methods.
 - Earlier detection of defects caused a **31%** decrease of rework efforts.
 - Elimination of defects prior to version release increased from 94% to **almost 100%**.
 - A **140%** increase in general productivity.

Quality management maturity grid

[P. Crosby – Quality is free]

Measurem. Categ.	Stage I Uncertainty	Stage II Awakening	Stage III Enlightenment	Stage IV Wisdom	Stage V Certainty
Manag. Underst. & attitude					
Quality org. status					
Problem handling					
Costs of quality as % of sales	20	18	12	8	2.5
Q. improv. actions	We don't know why we have problems with quality				We know why we don't have problems with quality
Company quality pos.					

ISO – CMM Differences

ISO9001:2000	CMMI-DEV
International standard, applies to all types of organizations, supports both product and service oriented organizations	Written specifically for software development companies
A brief document – about 25 pages long, identifying the minimal requirements for a quality system	A detailed document – over 500 pages long
Emphasizes on a management of continuous improvement process, based on the PDCA (Plan-Do-Check-Act) model	Emphasizes on achieving “maturity” and improving its process continuously
One level of standard. The standard is based on recommendation	Defines 5 maturity levels of the organization, covering 25 process areas (PAs)

ISO – CMM Differences

ISO 9000	SW-CMMI
Outwardly focused	Inwardly focused
Minimum requirements with implied continuous improvements	Explicit continuous quality improvement
Registration Document	No documentation

Certification audit for a 50 employee organization will be executed by 1 -12 auditors during one day	Certification audit for a 50 employee organization will be executed by 4 auditors during 4-5 days
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